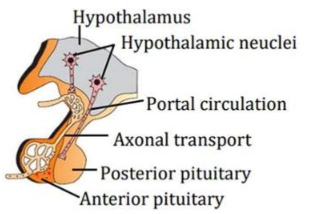
Chemical Coordination and Integration

1. **Given below are two statements :**
   Statement-I: Thyroid gland secretes a protein hormone called thyrocalcitonin (TCT) which regulates the blood calcium levels.
   Statement-II: The thyroid gland is composed of two lobes which are located on either side of the trachea.
   (1) Statement I and Statement II both are correct.
   (2) Statement I is correct, but statement II is incorrect.
   (3) Statement I is incorrect, but Statement II is correct.
   (4) Statement I and Statement II both are incorrect.
2. **GnRH, a hypothalamic hormone, needed in reproduction, acts on:**
   (1) anterior pituitary gland and stimulates secretion of LH and FSH.
   (2) posterior pituitary gland and stimulates secretion of oxytocin and FSH.
   (3) posterior pituitary gland and stimulates secretion of LH and relaxin.
   (4) anterior pituitary gland and stimulates secretion of LH and oxytocin.
3. **Hormones are;**
   (1) Nutrient chemicals
   (2) Intracellular messengers
   (3) Produced in trace amounts
   (4) Produced by exocrine glands
4. **Both adrenaline and cortisol are secreted in response to stress. Which of the following statements is also true for both of these hormones?**
   (1) They act to increase blood glucose
   (2) They are secreted by the adrenal cortex
   (3) Their secretion is stimulated by adrenocorticotropin
   (4) They are secreted into the blood within seconds of the onset of stress.
5. **Match List-I with List-II**

List-I	List-II
(a) Pituitary hormone	**(i) Steroid**
(b) Epinephrine	**(ii) Neuropeptides**
(c) Endorphins	**(iii) Peptides, proteins**
(d) Cortisol	**(iv) Biogenic amines**

   (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
   (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
   (3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
   (4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
6. **Which one is incorrect statement ?**
   (1) Hypothalamus regulate a wide spectrum of body functions.
   (2) Pituitary, pineal, testes, heart & kidney are organised endocrine gland of body.
   (3) Hormones are non-nutrient chemicals and intercellular messenger.
   (4) LH helps in maintenance of corpus luteum after ovulation.
7. **Find out incorrect labelling in following diagram**



   (1) Axonal transport and Anterior pituitary
   (2) Portal circulation and Posterior pituitary
   (3) Anterior pituitary and Posterior pituitary
   (4) Portal circulation and Axonal transport

8. **Select the correct pair.**
   (1) Low secretion of GH – Acromegaly
   (2) High secretion of GH - Pituitary dwarfism
   (3) LH in female - Induce ovulation
   (4) High secretion of ADH - Enhance loss of water from the body

9. **Read the following statements.**
   I. Adrenaline is also called epinephrine.
   II. Adrenaline and nor-adrenaline are collectively called catecholamines.
   III. Epinephrine and nor-epinephrine are released at the time of stress and emergency.
   IV. Insulin is a catecholamine.
   Choose the option representing correct statements.
   (1) I & IV
   (2) II, III & IV
   (3) I, II, III & IV
   (4) I, II & III

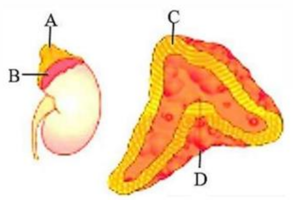
10. **Given below are two statements :**
    Assertion(A): The atrial wall of our heart secretes a very important peptide hormone called atrial natriuretic factor (ANF), which decreases blood pressure.
    Reason(R): ANF causes dilation of the blood vessels.
    In the light of the above statements, choose the most appropriate answer from option given below
    (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
    (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
    (3) Assertion (A) is true, and Reason (R) is false.
    (4) Assertion (A) is false, and Reason (R) is true.

11. **Select incorrect statement about hormonal feedback mechanism.**
    (1) Primarily maintained by hormones.
    (2) Positive feedback mechanism inhibits the secretion.
    (3) Negative feedback mechanism normalises the things when they start becoming extreme.
    (4) It is essential for maintaining body's equilibrium (homeostasis).

12. **Hypothyroidism from infancy leading to dwarfism and mental retardation is referred to as;**
    (1) Exophthalamic goitre
    (2) Cretinism
    (3) Simple goitre
    (4) Thyroid myxedema

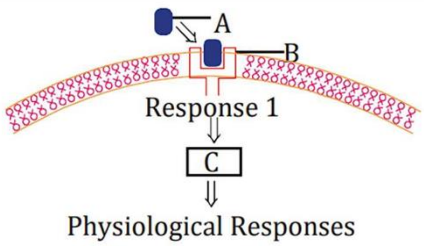
13. **Consider the following statements**
    Assertion(A): Iodine is essential for the normal rate of hormone synthesis in the thyroid.
    Reason(R): Deficiency of iodine in our diet results in hypothyroidism and enlargement of the thyroid gland.
    (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
    (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not correct explanation of Assertion (A).
    (3) Assertion (A) is true, and Reason (R) is false.
    (4) Assertion (A) is false, and Reason (R) is true.

14. **A, B, C and D labels in the given figure are correctly shown in;**



    (1) A- Adrenal gland, B- Fat, C-Cortex, D- Medulla
    (2) A- JGA, B- Fat, C- Cortex, D-Medulla
    (3) A- Adrenal gland, B- Fat, C-Medulla, D- Cortex
    (4) A- Adrenal gland, B- Fat, C- Pars distalis, D- Pars intermedia

15. Identify A, B and C in the diagrammatic representation of the mechanism of hormone action. Select correct option from the following :



(1) A-Steroid Hormone; B-Hormone receptor Complex, C-Protein

(2) A-Protein Hormone, B-Receptor; C Cyclic AMP

(3) A-Steroid Hormone; B-Receptor, C - Second Messenger

(4) A-Protein Hormone; B-Cyclic AMP, C Hormone-receptor Complex

16. Urinary excretion of Na+ is regulated by

(1) Anterior Pituitary

(2) Adrenal cortex

(3) Neurohyphophysis

(4) Pars intermedia

17. Identify the hormone with its correct matching of source and function :

(1) Oxytocin - posterior pituitary, growth and maintenance of mammary glands.

(2) Melatonin - pineal gland, regulates the normal rhythm of sleepwake cycle.

(3) Progesterone - corpus-luteum, stimulation of growth and activities of female secondary sex organs.

(4) Atrial natriuretic factor - ventricular wall increases the blood pressure.

18. Select among the following that have heterocrine nature.

(1) Thyroid and parathyroid

(2) Pineal gland and hypothalamus

(3) Gonads and pancreas

(4) Thymus and liver

19. Given below are two statements :

Assertion(A): The posterior pituitary is under the direct neural regulation of the hypothalamus.

Reason(R): Melatonin also influences metabolism, pigmentation, the menstrual cycle as well as our defense capability.

Choose the correct option from the options given below.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true, and Reason (R) is false.

(4) Assertion (A) is false, and Reason (R) is true.

20. Given below are two statements :

Statement-I: Glucagon reduces the cellular glucose uptake and utilisation.

Statement-II: Glucagon is a steroid hormone.

Choose the correct option from the options given below.

(1) Statement I and Statement II both are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

21. How many of the following is/are the effects of glucocorticoids?

a. Production of glucose from non-sugar precursors

b. Breakdown of triglycerides

c. Cellular uptake and utilisation of amino acids

d. Inhibition of inflammatory reactions

Select the correct option.

(1) One

(2) Three

(3) Two

(4) Four

22. Placenta also acts as an endocrine tissue and produces several hormones like

(1) LH, hPL and hCG

(2) Estrogen, progesterone, hPL and hCG

(3) Relaxin, hCG, hPL and thymosin

(4) Thyroxine, relaxin, hCG and hPL

23. Select the hormone which is not secreted by endocrine cells of GI tract

(1) Renin

(2) Gastrin

(3) Secretin

(4) Cholecystokinin

24. Choose the hormone which can enter inside the target cell.

(1) Thyroxine

(2) Thymosin

(3) Thyrocalcitonin

(4) Parathormone

25. Select the option that represents exclusively endocrine glands.

(1) Testes and thymus

(2) Parathyroid and pancreas

(3) Adrenal and parathyroid

(4) Ovary and adrenal

26. If by some means, the β-cells of pancreas get damaged, then the person may become/show the

(a) Hypoglycemic

(b) Increased glycogenesis in hepatocytes

(c) Hyperglycemic

(d) Decreased gluconeogenesis in hepatic cells

Choose the correct option from the options given below.

(1) (a) only

(2) (a) and (b)

(3) (c) only

(4) (c) and (d)

27. Hormone responsible for maintaining the normal sleep wake cycle is

(1) Serotonin

(2) Melatonin

(3) Oxytocin

(4) Vasopressin

28. If the thymus gland of a child gets damaged by any means, it will result in

(1) Low oxygen carrying capacity of haemoglobin

(2) Severe decrease in cell mediated immunity

(3) Loss of growth and development of bones

(4) Loss of water through urine

29. In case, there is an increase in the osmotic concentration of the blood plasma of a person, then, ADH can help in retention of water through interaction with target cells in the

a. Anterior pituitary

b. Urinary bladder

c. Posterior pituitary

d. DCT of nephron

e. Collecting duct

(1) a, d and e

(2) c and d

(3) b, c and d

(4) d and e

30. Which among following sets of hormones majorly helps a mother to produce & eject milk respectively

(1) Luteinizing hormone and estrogen

(2) Prolactin and FSH

(3) Prolactin and oxytocin

(4) FSH and luteinizing hormone

31. Trophic hormones from the anterior pituitary directly affect the release of which of the following hormones?

(1) Glucagon

(2) Melatonin

(3) Calcitonin

(4) Thyroxine

32. Match column I with column II w.r.t glands and the diseases associated with their hyposecretions.

Column I	Column II
a. Ovary	**(i) Simple goitre**
b. Adrenal	**(ii) Addison's disease**
c. Pancreas	**(iii) Osteoporosis**
d. Thyroid	**(iv) Diabetes mellitus**

(1) a(iii), b(i), c(iv), d(ii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(ii), b(i), c(iv), d(iii)

33. Injection of which of the following hormones is useful for the treatment of allergy?

(1) Aldosterone

(2) Cortisol

(3) Insulin

(4) ADH

34. Hormones which regulate pigmentation of the skin are secreted by

(1) Pineal gland and pituitary

(2) Thyroid and parathyroid

(3) Adrenal gland and Anterior pituitary

(4) Posterior pituitary and pineal gland

35. Hormones are produced in trace amounts as very few molecules of any hormone is required to produce changes in target cells".
Amongst the following, which one is the best possible explanation for the above stated fact?

(1) All hormones are lipid-soluble and readily penetrate the membrane of the target cells

(2) The mechanism of every hormone action involves the formation of hormone-receptor complex with the genome

(3) Hormones are large nutrient molecules that remain in blood circulation for a long time

(4) The mechanism of hormone action involves an enzyme cascade that amplifies the response of a hormone

36. Match the column-I with column-II

	Column-I		Column-II
I.	Decreased blood glucose levels	A.	Glycogenesis
II.	Process of conversion of glucose to glycogen	B.	Hypoglycaemia
III.	Hormones of fight and flight	C.	Addison's disease
IV.	Underproduction of hormones by the adrenal cortex	D.	Catechol amines

(1) I-A, II-B, III-D, IV-C

(2) I-C, II-D, III-B, IV-A

(3) I-B, II-A, III-D, IV-C

(4) I-C, II-D, III-A, IV-B

37. Which of the following is a sex hormone and also stimulates erythropoiesis?

(1) Thyroxine

(2) Estrogen

(3) Cortisol

(4) Testosterone

38. Assertion (A): At the effector sites, the response produced by the steroidal epinephrine is faster as compared to the effect shown by MSH.
Reason (R): Protein hormones interact with receptors that are present on the nucleus of the cell whereas steroidal hormones typically interact with receptors present on the surface of the cell membrane.
Choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true; (R) is false

39. Choose the incorrect statement w.r.t humans.

(1) Neurohypophysis releases ADH and oxytocin.

(2) The pituitary hormones regulate the growth and development of somatic tissues and the activities of some peripheral endocrine glands.

(3) Gastrin and GIP hormone show synergistic effects

(4) Responses of the endocrine system are often slower than the responses of the nervous system.

40. Read the following statements w.r.t humans:

Statement A: Insulin mainly acts on hepatocytes and adipocytes, and enhances cellular glucose uptake and utilisation.

Statement B: The specific action of PTH is to decrease the amount of calcium in blood.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

41. All of the following are true w.r.t aldosterone, except

(1) Regulates the homeostasis of Na+ and K+ in body fluid

(2) Is released by the innermost layer of adrenal gland called adrenal medulla

(3) Helps to regulate blood pressure and blood volume PSYCO

(4) Maintains ionic balance in the body

42. In human females, menstrual cycle is affected by

a. Thyroxine

b. Melatonin

c. Estrogen

d. Prolactin

Choose the correct option.

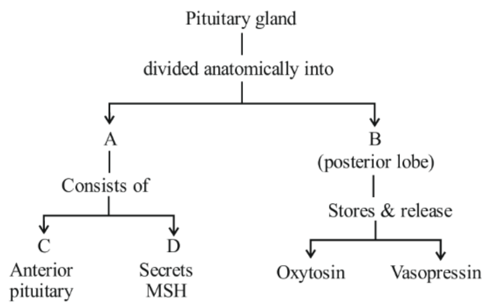
(1) c and d only

(2) a, b and c only

(3) a and c only

(4) a, b, c and d

43. Identify A, B,C and D :



(1) A-Neurohypophysis, B-Adenohypophysis, C Pars distalis, D-Pars intermedia

(2) A-Adenohypophysis, B-Neurohypophysis, C Pars intermedia, D-Pars distalis

(3) A-Adenohypophysis, B-Neurohypophysis, C Pars distalis, D-Pars intermedia

(4) A-Neurohypophysis, B-Adenohypophysis, C Pars intermedia, D-Pars distalis

44. The functioning of adrenal medulla gland is similar to those of nerves because :

(1) Adrenal medulla and nervous system are derived from embryonic mesoderm

(2) Adrenal medulla and nerves secrete similar chemicals and adrenaline and nor-adrenaline

(3) Adrenal medulla does not secrete any hormone

(4) Adrenal medulla is made up of nervous tissue

45. Which statements is false ?

(a) Steroid hormones act by passing through the cell membrane and affecting gene transcription

(b) Adrenaline (epinephrine) acts on cell membrane receptors to stimulate production of second messengers

(c) The anterior pituitary releases several hormones which act on other endocrine glands

(d) Insulin is produced in the islets of langerhans in the spleen when blood sugar levels fall

(1) a and b

(2) b only

(3) c and d

(4) d only

46. Which of the following structures which are associated with endocrine system but do not have its homology with nervous system :

(i) Hypothalamus

(ii) Adrenal cortex

(iii) Adrenal medulla

(iv) Pituitary

(v) Thyroid

(1) i and ii

(2) ii and v

(3) ii and iii

(4) iii and iv

47. Given below are two statements :

Assertion : Other then gland some other organs, e.g., gastrointestinal tract, liver, kidney, heart also produce hormones.

Reason : Hypothalamic hormone called Gonadotrophin releasing hormone (GnRH)

(1) Both Assertion & Reason are True and the Reason is the correct explanation of the Assertion

(2) Both Assertion & Reason are True but the Reason is not correct explanation of the Assertion

(3) Assertion is True statement but Reason is false

(4) Both Assertion and Reason are False statements

48. Which of the following statements is concerned with endocrine glands and their secretions :

(1) Endocrine glands lack ducts and are hence called ductless glands

(2) Chemical produced by endocrine glands and released into blood and transported to a distantly located target organ is the classical definition of hormones

(3) Hormones are non-nutrient chemical which act as intercellular messengers and are produced in trace amount according to current scientific definition

(4) All of the above

49. Identify the hormone with its correct matching of source and function.

(1) Oxytocin- Posterior pituitary, growth and maintenance of mammary glands

(2) Melatonin - Pineal gland, regulates the normal rhythm of sleep wake cycle

(3) Progesterone - Corpus luteum, stimulation of growth and activities of female secondary sex organs

(4) Atrial natriuretic factor - Ventricular wall increases the blood pressure.

50. Select the names of the hormones to which the following terms are associated :

A. Biological clock

B. Pregnancy hormone

C. Hypercalcaemic hormone

D. Anti diabetic hormone

E. Milk let down factor

(1) Melatonin, Oestrogens, Calcitonin, Glucagon, Oxytocin

(2) M.S.H., Progesterone, Parathormone, Insulin, Oxytocin

(3) Melatonin, Progesterone, Parathormone , Insulin

(4) M.S.H., Oestrogens, Calcitonin, Glucagon, Oxytocin

Answer Key

1. (1)
2. (1)
3. (3)
4. (1)
5. (2)
6. (2)
7. (4)
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50. (3)